

SHALINI BANSAL

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SUMMARY

Mechanical engineer with 4+ years in mechanism design, concept-to-production development, and hardware and software validation on CNC machines. Experienced in leading design reviews and owning the life cycle of a product. Hands-on experience with precision machining and electromechanical prototyping, with a strong drive to invent and design high-performance mechanical systems.

EDUCATION

Master of Science in Computer Science , <i>Sofia University, Irvine, California</i>	<i>Jan 2025 – Present</i>
Master of Science in Mechanical Engineering , <i>University of California, Riverside, California</i>	<i>Jan 2021 – Dec 2021</i>
Bachelor of Engineering in Mechanical Engineering , <i>Visvesvaraya Technological University, India</i>	<i>Aug 2016 – Sep 2020</i>

WORK EXPERIENCE

Associate Mechanical Engineer, *Haas Automation Inc., Oxnard, CA* *March 2026 – Present*

- Led end-to-end redesign of chip conveyor mechanisms inherited mid-cycle. Evaluated designs, identified possible failures, and drove assembly to production-ready specs through iterative design and DFM review.
- Redesigned coolant sanitizer assembly from an inherited NPI concept, scoped requirements, performed root cause analysis on leakage failures, and iterated on mechanism design to meet manufacturing and usability targets.
- Contributed to cross-functional design reviews, driving design decisions across the mechanical engineering department.
- Presented chip conveyor design and application guidance at the Haas Dealer Meeting, directly influencing purchasing decisions across dealer networks.

Machine Test Engineer 2, *Haas Automation Inc., Oxnard, CA* *Jun 2022 – March 2026*

- Designed test components in Fusion 360, generated G-code, and machined these parts on CNC machines to validate software features against engineering requirements.
- Led hardware integration on R&D machines, translated mechanical constraints into software requirements, coordinated across teams, and validated performance of CNC machine motion through live testing.
- Drove root cause analysis on software bugs by machining on CNC machines and used DFM/DFA principles to identify failure modes and verify fixes through structured test plans.
- Automated 200+ manual regression tests using Python, reducing manual testing overhead, and enabling biweekly automated validation of CNC software features.

Graduate Research Assistant, *University of California, Riverside, CA* *Jan 2021 – Dec 2021*

- Designed torsion-actuator assembly components in SolidWorks for a steerable robotic catheter, taking the mechanism from concept through prototype, applying GD&T and DFM throughout.
- Fabricated prototypes using SLA resin 3D printing and conducted motion tests on electromechanical assemblies, implementing design changes to improve steering performance and structural integrity.

Mechanical Engineering Intern, *Hindustan Aeronautics Limited, India* *Feb – Jun 2020*

- Performed FEA on an aircraft engine blisk to optimize machining parameters, reducing cycle time by 6% on a 5-axis CNC mill; verified surface roughness of machined prototype against engineering specs.

TECHNICAL SKILLS

CAD/CAE: SolidWorks, CATIA V5, Siemens NX, Fusion 360, AutoCAD, FEA, Topology Optimization, GD&T (ASME Y14.5)

Manufacturing/Machining: CNC Machining, Sheet Metal, 3D Printing, Casting, Composites, DFM/DFA, Lean Manufacturing

Software: Python, SAP, ECTR, TargetProcess, Jira, Perforce

LEADERSHIP & COMMUNITY

- Designed and ran a training program for all incoming engineers, covering machining workflows and hands-on CNC machining.
- Represented Gene Haas Foundation at the FIRST Robotics World Championship, leading technical demonstrations of CNC manufacturing technology and engaging with student engineering teams.
- Mentored two undergraduates on graduate school applications, research direction, and career planning; led onboarding discussions for 300+ incoming undergraduates at UCR.